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*Go, change the world*

**Theme: ENERGY**

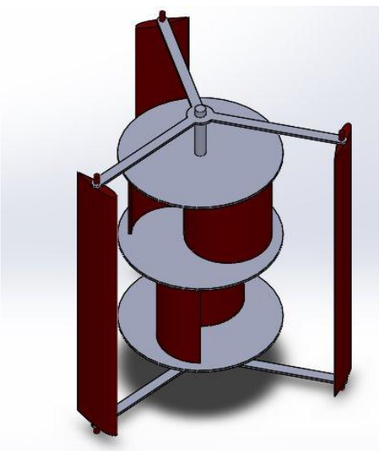
## HYBRID WIND AND SOLAR POWER GENERATOR WITH DATA ANALYSIS

## Introduction

The hybrid VAWT and solar energy generator aims to address the dual problems of space limitations and energy needs. When these hybrid systems are placed in between the street lights of road dividers, they allow sustainable energy generation, using both wind from passing vehicles as well as sunlight.

## Problem Definition

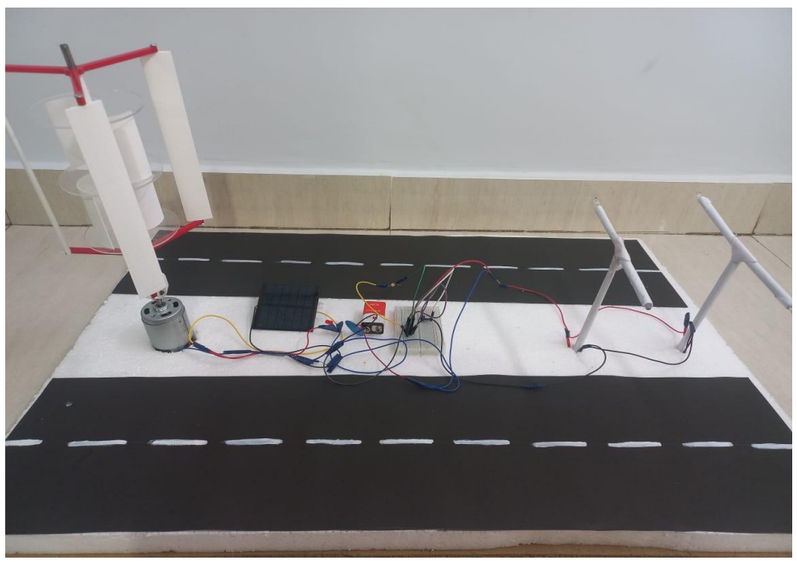
The increasing demand for renewable energy solutions highlights the need for efficient and sustainable systems. The current ones are very expensive, inefficient and cannot be used in urban areas due to size issues. Our Hybrid generator can solve these issues by maximizing energy outputs, enabling usage in urban areas, reduce manufacturing expenses and maintenance costs.



## Objectives

1) Develop a hybrid system, that can generate dual power with VAWT and solar panels.
2) Minimize space utilization, by designing space efficient Generator.
3) Enable constant power generation in all weather conditions.

## Methodology



**Phase-1**
- Objective.
- Problem Definition.
- Literature review.

**Phase-2**
- List of Components Used.
- Data Interpretation from previous weather patterns.
- Individual analysis of Darrieus and Savonius Turbines.

**Phase-3**
- 3-D Printing of VAWT.
- Assembly of components and connections.
- Model preparation.

## Tools used

1. **SolidWorks**: Design of Darrieus turbine, Savonius turbine & Prototype Assembly.
2. **Merra**: Plotting graph of Max & Min wind Speed over a specified time range.
3. **MSN-Weather**: Wind direction & speed over a specific area, Cross winds & Weather Reports.
4. **PVGIS 5.3**: Energy output from Solar Energy.
5. **Omni-Calculator**: Energy output from the wind.
6. **RVCE-Knimbus**: Research Paper & Documents for understanding.

## Results and Discussions

1) **Urban Adaptability and Space Efficiency:**
The compact design of the system makes it ideal for urban areas, fitting into small spaces like road dividers while keeping noise levels low.

2) **Cost-Effectiveness and Sustainability:**
The system offers long-term cost savings and sustainability, with low maintenance and reduced environmental impact over time.

3) **System Performance under Varied Weather Conditions:**
The hybrid system performs consistently across different weather scenarios, with solar energy boosting output on sunny days and wind power filling in during cloudy or night time.

4) **Public Acceptance:**
The compact and low-noise design made the system more acceptable for urban residents, with positive feedback regarding its aesthetic appeal and functionality.

## Conclusions

The hybrid solar and wind energy generation system demonstrates a sustainable and efficient approach to addressing the growing energy needs of urban and rural communities, by combining the strengths of solar panels and Vertical Axis Wind Turbines (VAWTs). This project offers a reliable, decentralized power solution that mitigates the challenges of intermittency in renewable energy.

**Key Points taken Away**:

I. Understanding Hybrid Renewable Energy Systems
II. Vertical Axis Wind Turbines (VAWTs)
III. Solar Energy Integration and Streetlight Applications.

## References


1) Variable design of Vertical Axis wind turbine - Ken-Yeen Lee , Andrew Cruden , Jo-Han Ng and Kok-Hoe Wong. July 2023.
2) **Recent Progress in Design and Performance Analysis of Vertical-Axis Wind Turbines—A Comprehensive Review -** Published: 27 May 2024
3) Wind turbine technology - fundamental concepts of wind turbine engineering, 2nd Edition(July)(2022)
4) *Wind Turbines: Fundamentals, Technologies, Application, Economics*; Springer: Berlin/Heidelberg, Germany, (December)(2021)
5) ) Global wind energy council report 2018, Wind Global Council Energy (April) (2023)


## QR Code of the Demonstration Video